

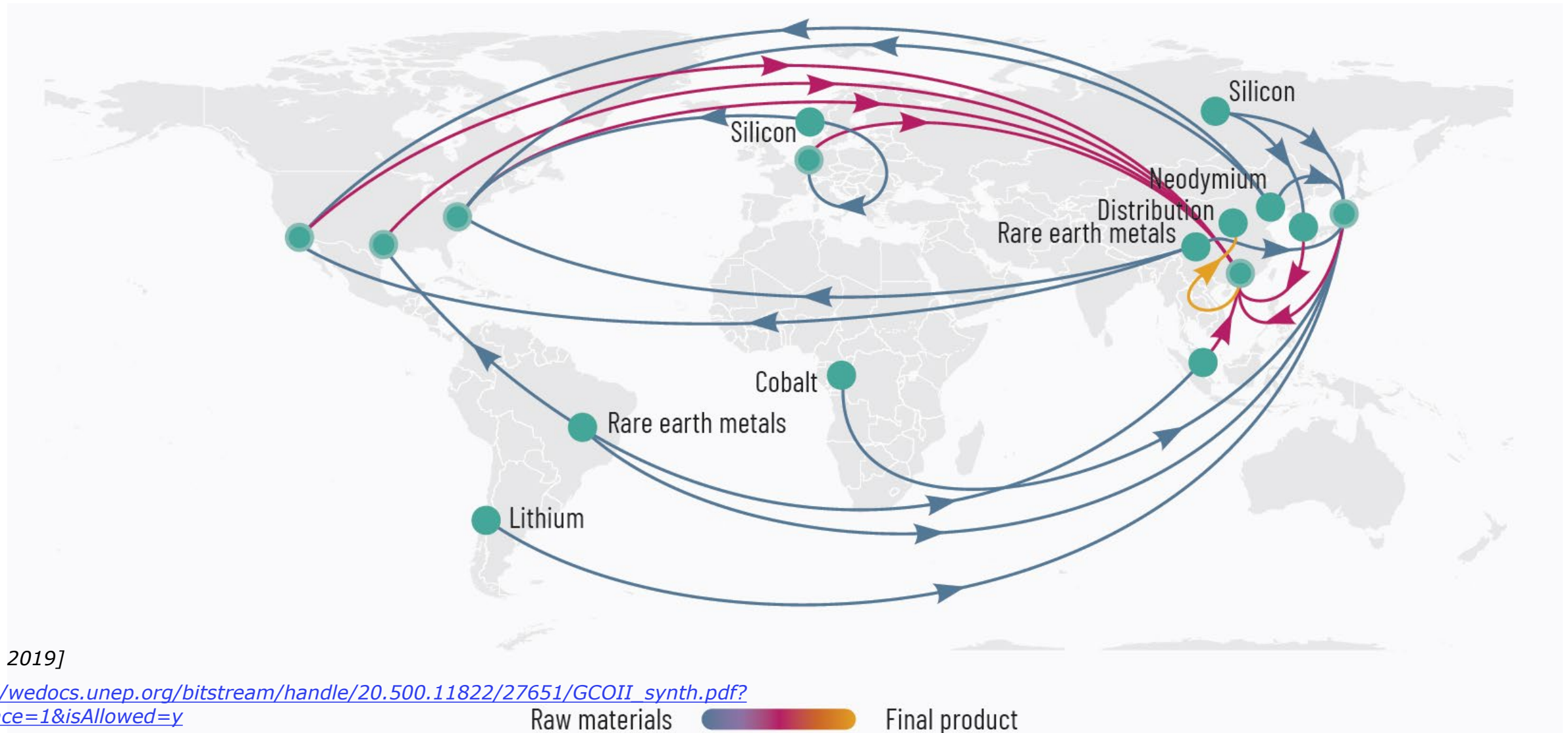
Stig Irving Olsen, Associate Professor

Quantitative Sustainability: The assessment framework

Learning outcomes of video

- Illustrate the generic life cycle of products and systems
- Explain the fundamentals of LCA
- Identify indicators for life cycle sustainability assessment
- Formulate a simplified assessment approach

Value chain is the life cycle - from cradle to grave





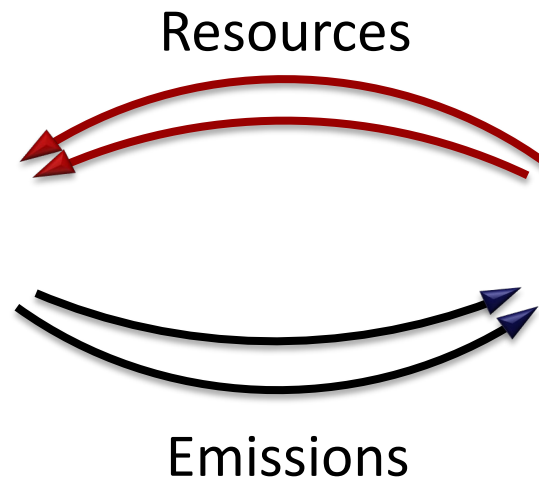
What is LCA?

How we understand the life cycle - from cradle to grave

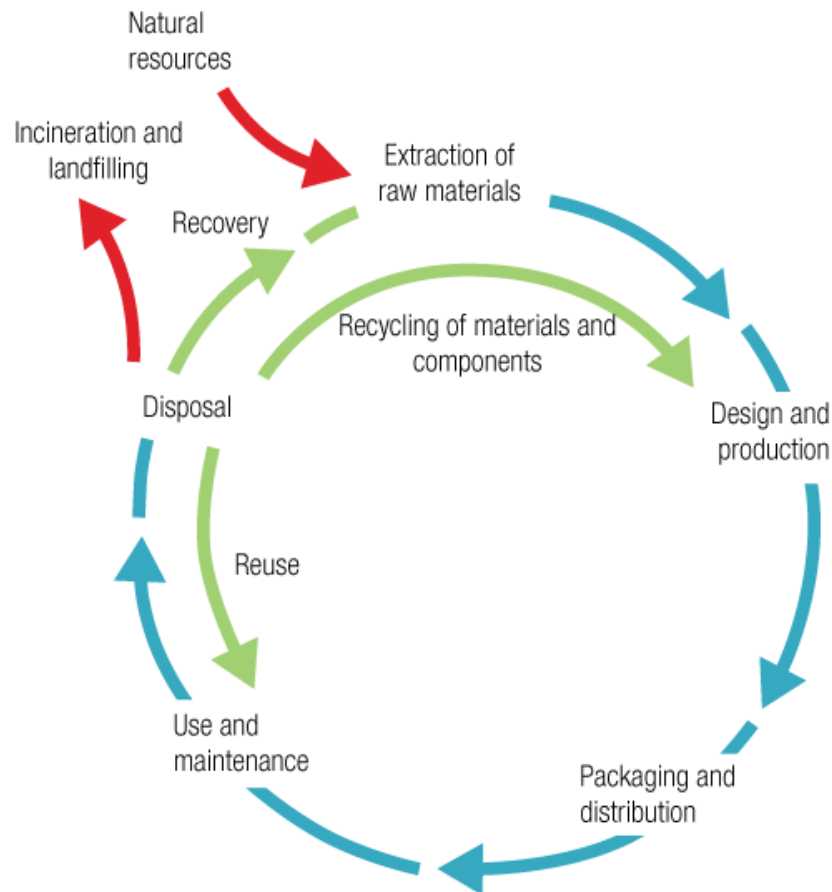
Activity



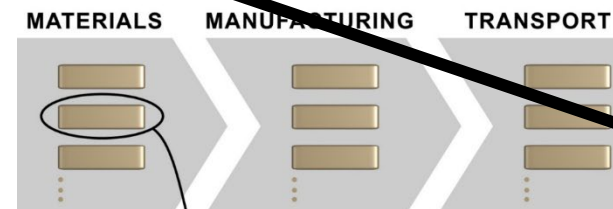
Technosphere
(Production system)



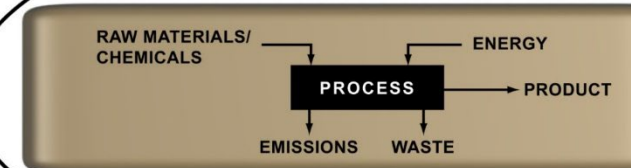
Ecosphere



PRODUCT SYSTEM



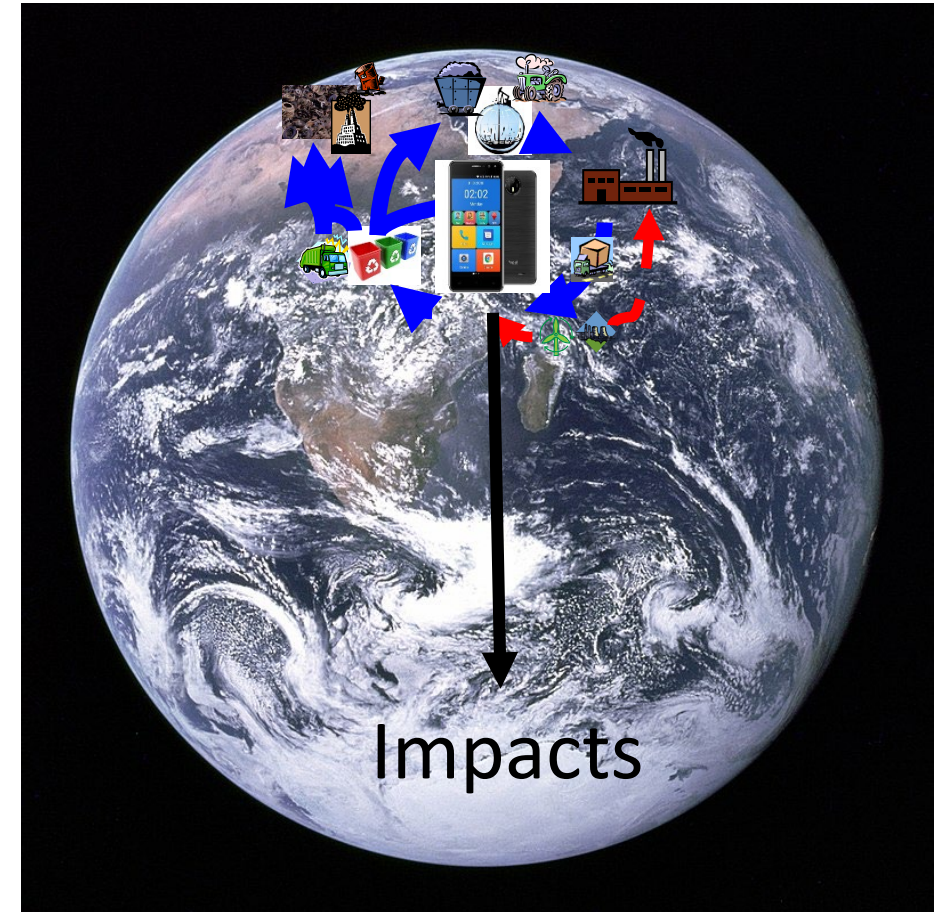
THE PROCESS AS THE FUNDAMENTAL ELEMENT OF THE PRODUCT SYSTEM



Substance	CAS.no.	Emission to air g	Emission to water g
2-hydroxy-ethanacrylate	816-61-0	0,0348	
4,4-methylenebis cyclohexylamine	1761-71-2	5,9E-02	
Ammonia	7664-81-7	3,7E-05	4,2E-05
Arsenic (As)	7440-38-2	2,0E-06	
Benzene	71-43-2 (cur)	5,0E-02	
Lead (Pb)	7439-92-1	8,5E-06	
Butoxyethanol	111-76-2	6,6E-01	
Carbondioxide	124-38-9	2,6E+02	
Carbonmonoxide (CO)	630-08-0	1,9E-01	
Cadmium (Cd)	7440-46-9	2,2E-07	
Chlorine (Cl ₂)	7782-50-5	4,6E-04	
Chromium (Cr VI)	7440-47-3	5,3E-06	
Dicyclohexane methane	86-73-6	5,1E-02	
Nitrous oxide(N ₂ O)	10024-97-2	1,7E-02	
2,4-Dinitrotoluene	121-14-2	9,5E-02	
HMDI	5124-30-1	7,5E-02	
Hydro carbons (electricity, stationary combustio	-	1,7E+00	
Hydrogen ions (H ⁺)	-	-	1,0E-03
i-butanol	78-83-1	3,5E-02	
i-propanol	67-63-0	9,2E-01	
copper (Cu)	7740-50-8	1,8E-05	
Mercury(Hg)	7439-97-6	2,7E-06	
Methane	74-82-8	5,0E-03	
Methyl i-butyl ketone	108-10-1	5,7E-02	
Monoethyl amine	75-04-7	-	7,9E-06
Nickel (Ni)	7440-02-0	1,1E-05	
Nitrogen oxide (NO _x)	10102-44-0	1,1E+00	
NM/OC, diesel engine (exhaust)	-	3,9E-02	
NM/OC, power plants (stationary combustion)	-	3,9E-03	
Ozone (O ₃)	10028-15-6	1,8E-03	
PAH	ikke specifik	2,4E-08	
Phenol	108-95-2	-	1,3E-05
Phosgene	75-44-5	1,4E-01	
Polyester polyo	ikke specifik	1,6E-01	
1,2-propylenoxide	75-56-9	8,2E-02	
Nitric acid	7782-77-6 (c	8,5E-02	
Hydrochloric acid	7647-01-0 (c	1,9E-02	
Selenium (Se)	7782-49-2	2,6E-05	
Sulphur dioxide(SO ₂)	7446-09-5	1,3E+00	
Toluene	108-88-3	4,8E-02	
Toluene-2,4-diamine	95-80-7	7,9E-02	
Toluene diisocyanat (TDI)	26471-62-5	1,6E-01	
Total-N	-	-	2,6E-05
Triethylamine	121-44-8	1,6E-01	
Unspecified aldehydes	-	7,5E-04	
Unspecified organic compounds	-	1,5E-03	
Vanadium	7440-62-2	1,8E-04	
VOC, diesel engine (exhaust)	-	6,4E-05	
VOC, stationary combustion (coal fired)	-	4,0E-05	
VOC, stationary combustion (natural gas fired)	-	2,2E-03	
VOC, stationary combustion (oil fired)	-	1,4E-04	
Xylene	1330-20-7	1,4E-01	
Zinc (Zn)	7440-66-6	8,9E-05	

What is LCA?

- Definition of goal of assessment - “what is the question?”
- Scoping of system
- Collection of data on emissions and resource use
- Translation of emissions into environmental impacts
- Interpretation of results - answer to the question



A fundamental feature of LCA is the focus on service



+ washing
& transport

Vs
Same
function



LCA is for comparisons



Is better than



But is it OK for the environment?
(Absolute perspective)

LCA helps avoiding problem shifting

- life cycle from cradle to grave
- All relevant environmental impacts
- working environment
- resource consumption (biotic and abiotic)

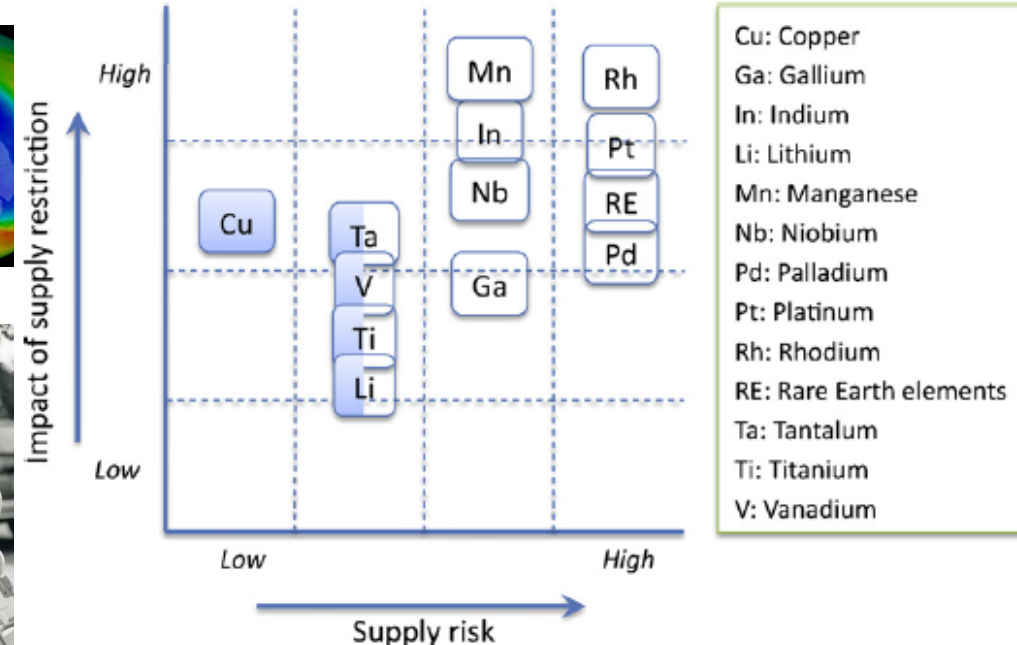
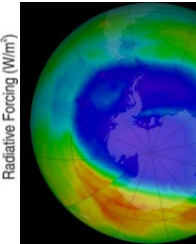
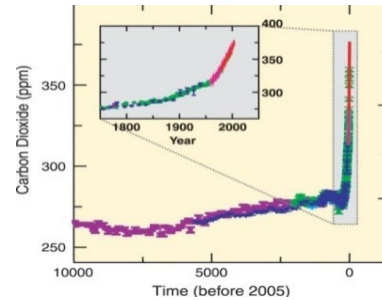
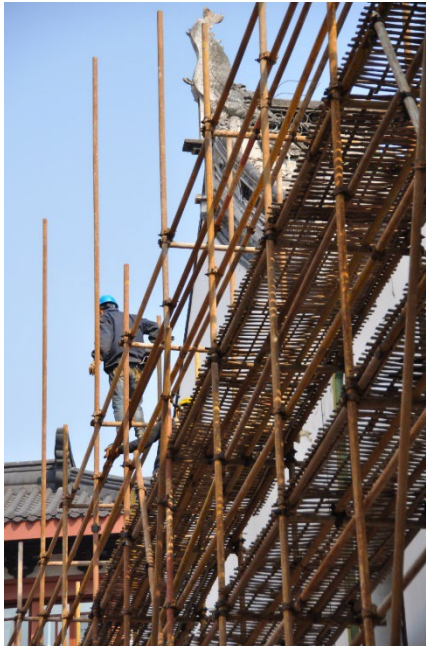
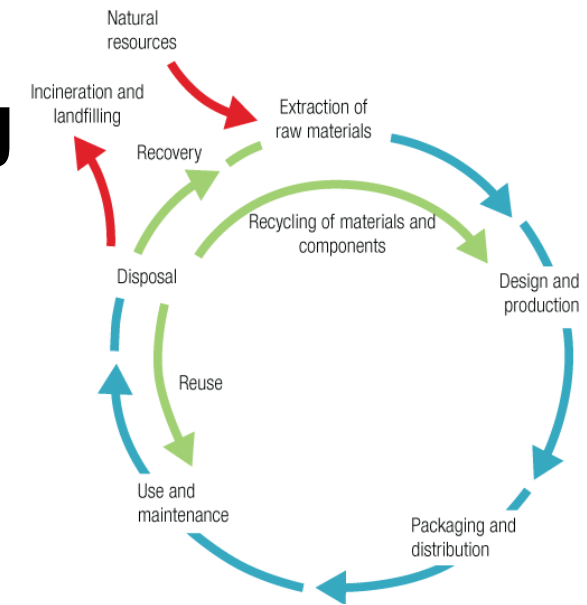
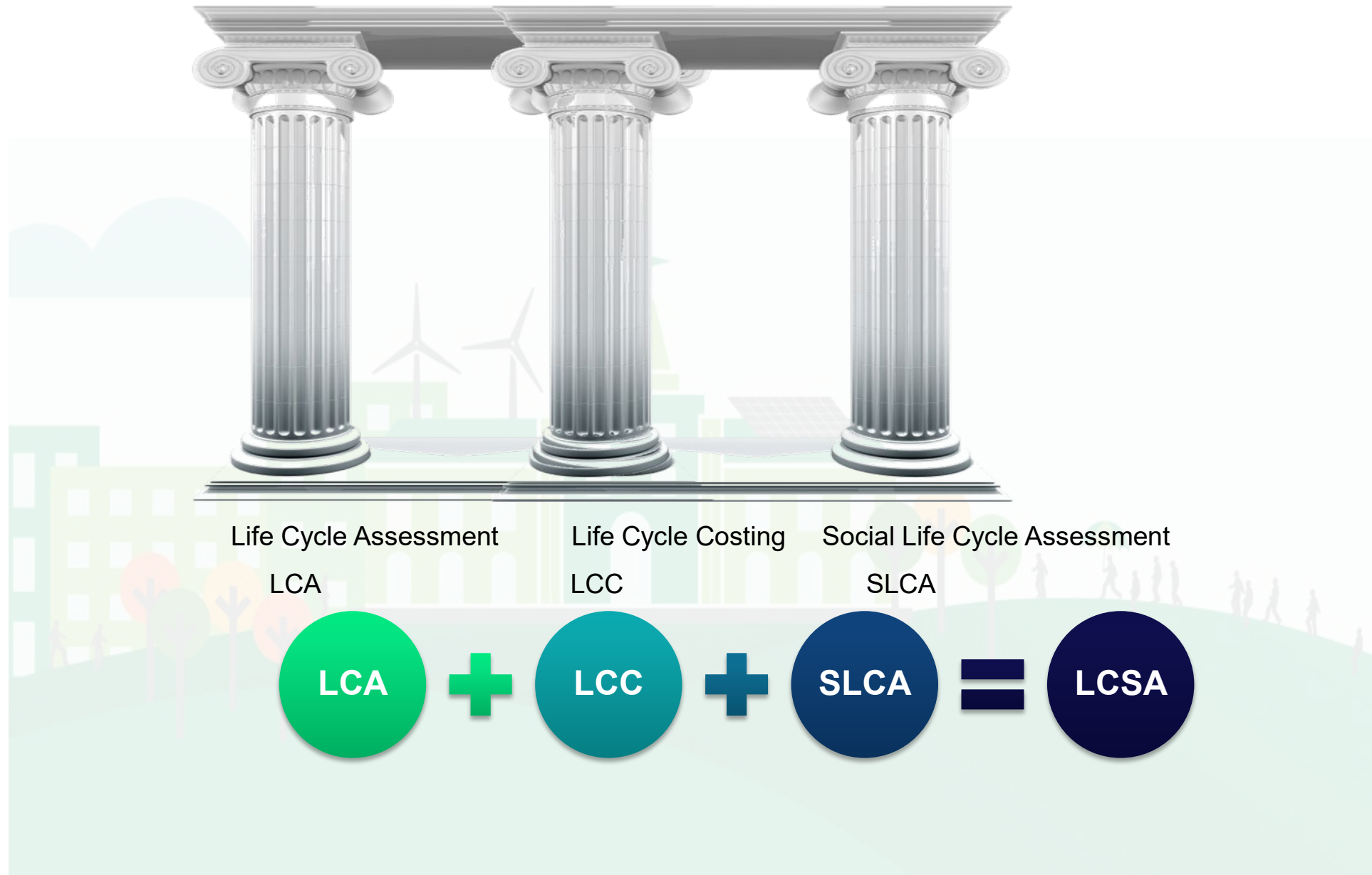
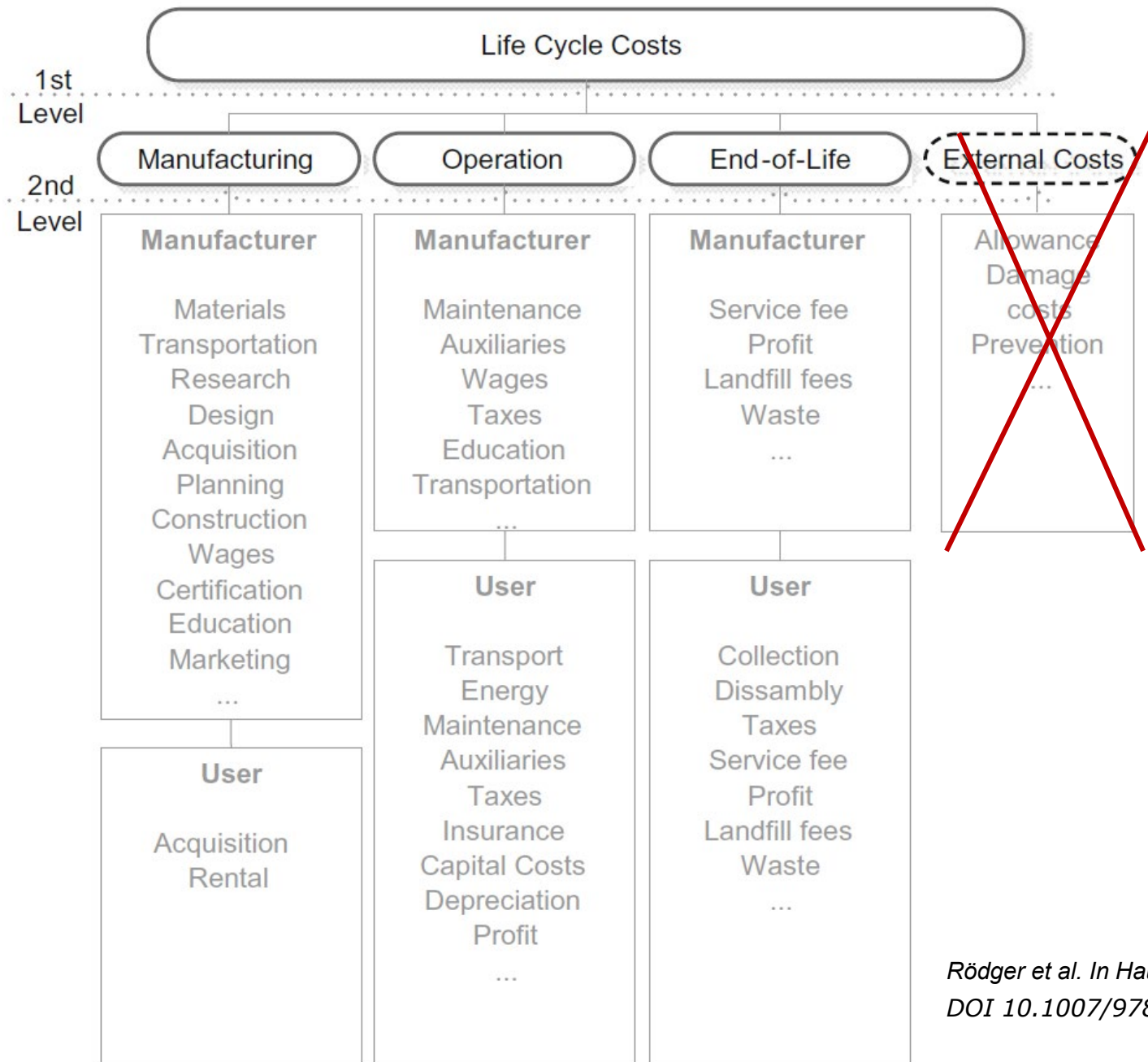


Fig. 2.3. Criticality grid for 11 materials.

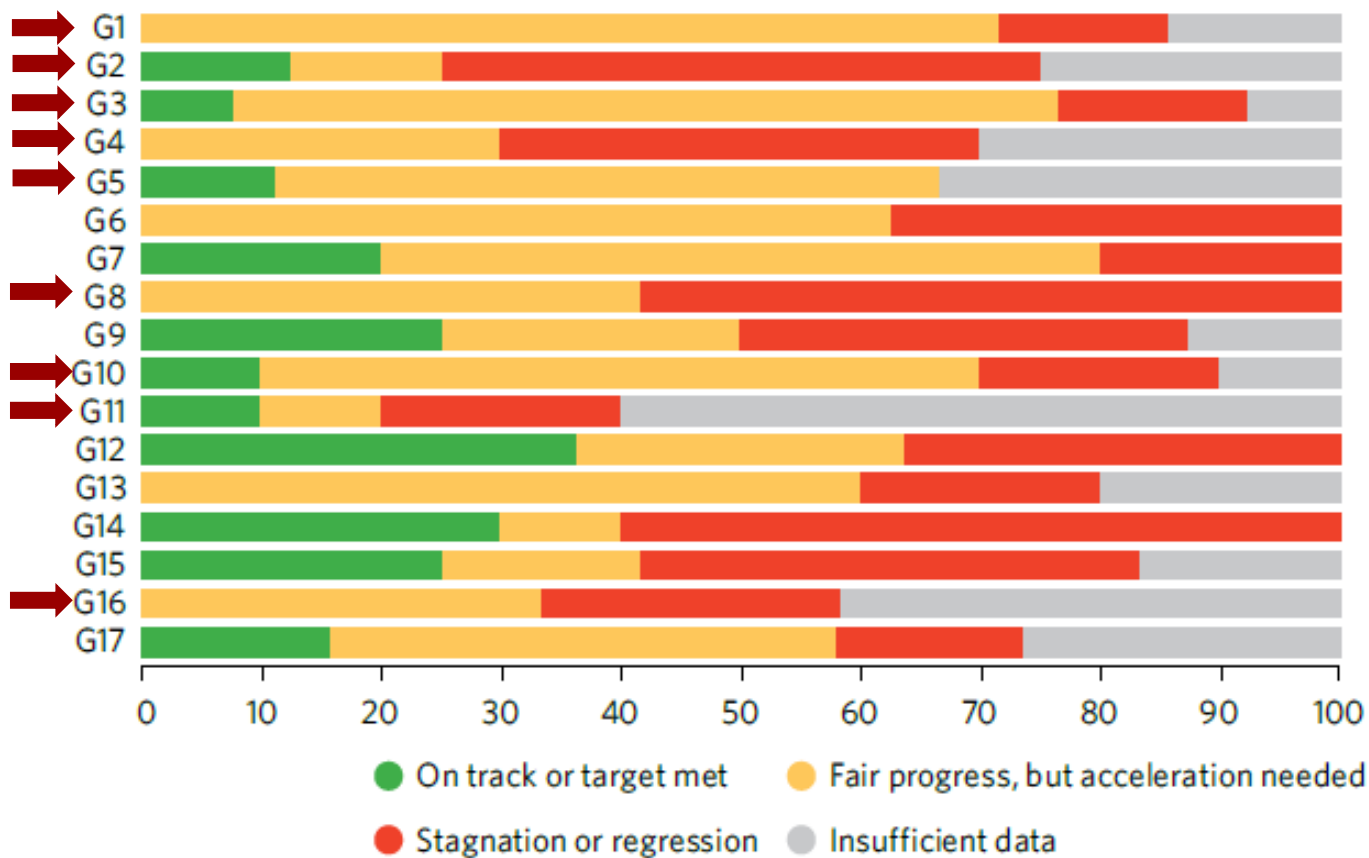
Life Cycle Sustainability Assessment (LCSA)





Rödger et al. In Hauschild et al (Eds) Life cycle Assessment, 2018
 DOI 10.1007/978-3-319-56475-3_15

The social side



<https://unstats.un.org/sdgs/report/2023/progress-midpoint/>

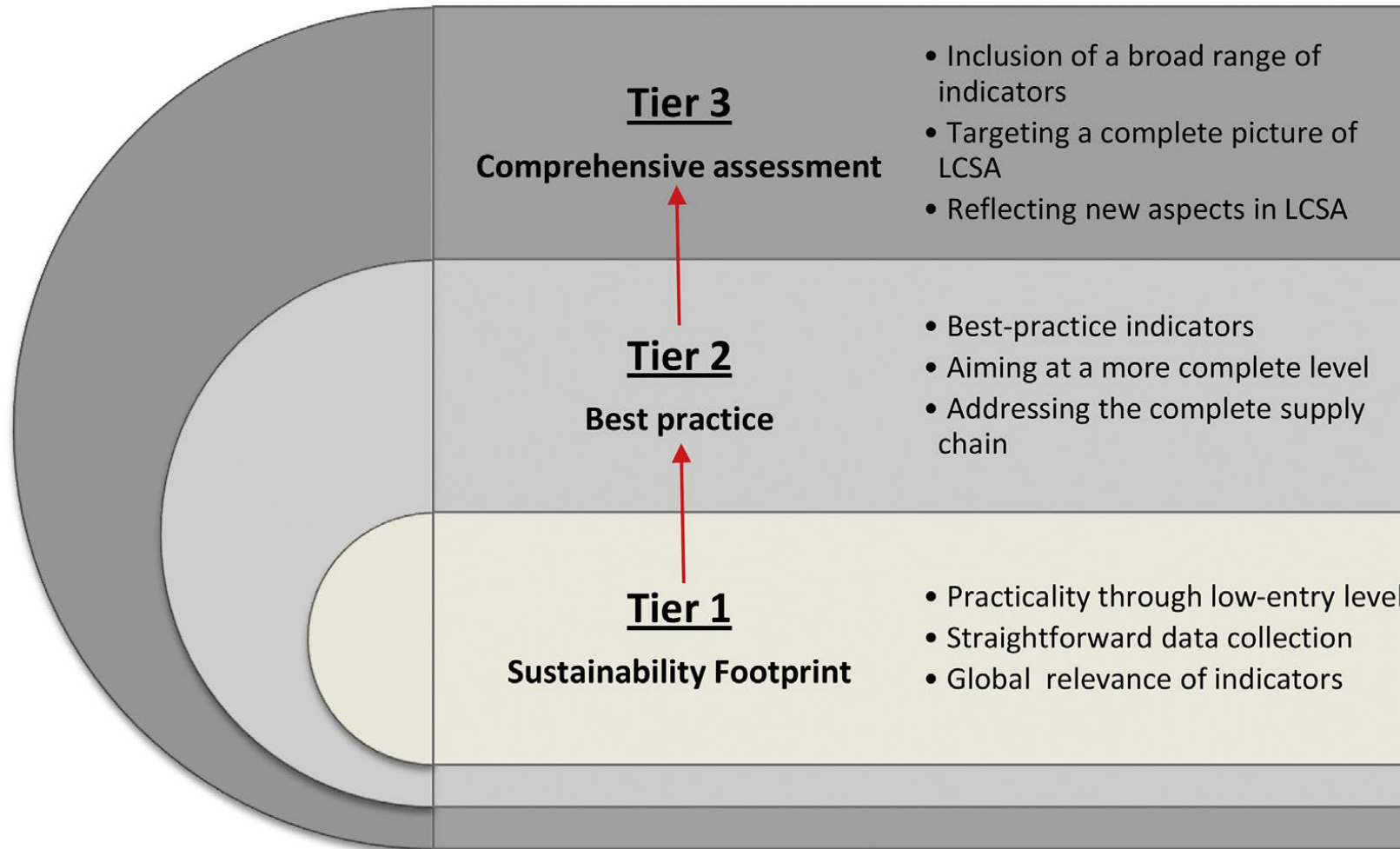


Stakeholder categories	Worker	Local community	Value chain actors (not including consumers)	Consumer	Society	Children
Subcategories	1. Freedom of association and collective bargaining 2. Child labor 3. Fair salary 4. Working hours 5. Forced labor 6. Equal opportunities / discrimination 7. Health and safety 8. Social benefits / social security 9. Employment relationship 10. Sexual harassment 11. Smallholders including farmers	1. Access to material resources 2. Access to immaterial resources 3. Delocalization and migration 4. Cultural heritage 5. Safe and healthy living conditions 6. Respect of indigenous rights 7. Community engagement 8. Local employment 9. Secure living conditions	1. Fair competition 2. Promoting social responsibility 3. Supplier relationships 4. Respect of intellectual property rights 5. Wealth distribution	1. Health and safety 2. Feedback mechanism 3. Consumer privacy 4. Transparency 5. End-of-life responsibility	1. Public commitments to sustainability issues 2. Contribution to economic development 3. Prevention and mitigation of armed conflicts 4. Technology development 5. Corruption 6. Ethical treatment of animals 7. Poverty alleviation	1. Education provided in the local community 2. Health issues for children as consumers 3. Children concerns regarding marketing practices

UNEP, 2020. Guidelines for Social Life

Cycle Assessment of Products and Organizations 2020. Benoît Norris, C., Traverso, M., Neugebauer, S., Ekener, E., Schaubroeck, T., Russo Garrido, S., Berger, M., Valdivia, S., Lehmann, A., Finkbeiner, M., Arcese, G. (eds.). United Nations Environment Programme (UNEP).

Stepwise Life Cycle Sustainability Assessment



Neugebauer et al, *Journal of Cleaner Production* 102 (2015)
<http://dx.doi.org/10.1016/j.jclepro.2015.04.053>

The MECO framework

	Life Cycle Stage				
Causes of Environmental Impact	Extraction of raw materials	Manufacturing stage	Use stage	Disposal stage	Driver for:
Materials					Depletion of natural resources
Energy					Climate change, acidification, photochemical ozone formation etc.
Chemicals					Human and ecological toxicity Ozone depletion
Others					E.g. land use, water use, social impacts etc.

Quantitative Sustainability framework

	Life Cycle Stage				
Sustainability Impact area	Extraction of raw materials	Manufacturing stage	Use stage	Disposal stage	Measured by:
Resources					Use of biotic and abiotic resources Circular economy indicators
Environment					Climate change, Carbon footprint Absolute boundaries
Economic					Costs
Social/Health					Socioeconomic impacts, health impacts
Transition					Qualitative or semiquantitative assessment

Summary

- LCA is a comparative method that assess the environmental impacts of a product or system throughout its life cycle
- It is important to precisely define the object of assessment
- Life Cycle Sustainability Assessment aims to additionally include the economic and the social impacts
- In this course we do not apply LCSA, but aim to cover all aspects in a life cycle perspective through a simplified approach